

Issues — Open Workable Items

Revision: 34 **Last modified:** 2026-08-21T18:38:54Z **Ticket prefix:** BOB (operator-mandated, 2026-06-06) **Scope:** Open/active items only. Closed items migrate to [Fixed.md](#).

Tracking: this file + [Issues_Summary.md](#) are authoritative for open work. The SQLite single-source-of-truth + docs_chain engine (BOB-010) is complete.

BOB-008 — RuTracker automated login blocked by CAPTCHA

Status: Operator-blocked **Type:** Bug **Created:** 2026-06-06
Operator-Block-Details: WHAT — RuTracker login with stored creds returns no session cookie (CAPTCHA wall). WHY — automated user/pass login is CAPTCHA-gated; self-resolution exhausted (creds correct + wired, login attempted, auth=True). UNBLOCK — [A] operator completes the CAPTCHA flow at /api/v1/auth/rutracker/captcha + /login. [B] operator pastes a fresh bb_session cookie via /auth/rutracker/cookie-login. WHO — operator.

Evidence: live search per-tracker stat rutracker status=error auth=True. The diagnostic now reports error_type="upstream_captcha" with the FACT-based message "rutracker login.php returned no session cookie — this is the rutracker anti-abuse CAPTCHA wall (gates login.php when logins spike), not a credential failure. Set RUTRACKER_COOKIES from a logged-in browser session to bypass the login round-trip." (download-proxy/src/merge_service/search.py). The earlier quote here — error="login returned no session cookie — likely CAPTCHA" — was SUPERSEDED when BOB-117 landed and no longer exists in source; it is corrected rather than left as stale evidence (§11.4.6 no-guessing, §11.4.7 evidence must reflect the conditions it claims).

BOB-065 — Lava P2: Egress diagnosis and VPN-host SOCKS routing (containers pkg/egress)

Status: Queued **Type:** Task **Severity:** High

[Backfill from RD2-15/GA-05, audit doc 2026-08-08] Lava-porting finding P2 (Egress decision + VPN-host routing, Lava PLAYBOOK sections 0 and 4). Problem Boba-Base shares: on a datacenter host, trackers are network-blocked (DNS-fail/TLS-MITM, not Cloudflare so FlareSolvrr cannot fix). Affects Jackett indexer fetches + merge_service/download-proxy + plugin engines. Diagnosis (port the script): curl https://api.ipify.org (host IP) + curl -o /dev/null -w %{http_code} https:// direct vs via a VPN-host SOCKS proxy. Different egress IP + 200 via proxy confirms. Fix: route outbound through a VPN-connected host (the nezha pattern). SOCKS tunnel ssh -D 127.0.0.1:1080 -N (use -socks5-hostname for remote DNS); point Jackett + download-proxy + qBitTorrent-go at it (P3). For browser-cookie harvest, run the harvester ON the VPN host. Port: containers submodule pkg/egress (tunnel up/verify) + scripts/egress-via-vpn.sh glue; reuse Boba-Base existing ensure-macos-tunnel.sh style. TDD: assert the via-proxy egress IP != direct host IP AND a known-blocked tracker returns 200 via proxy. Source: docs/PORTING-FROM-LAVA.md. Per audit RD2-15 [P0]: Create tracked workable items (BOB-064..067) for the four Lava-porting findings, citing implementing commits as evidence, closed as Implemented.

BOB-066 — Lava P3: BOBA_UPSTREAM_PROXY in download-proxy + qBitTorrent-go + Jackett + compose env-forward

Status: In progress **Type:** Task **Severity:** High

[Backfill from RD2-15/GA-05, audit doc 2026-08-08] Lava-porting finding P3 (Configurable outbound proxy in the services, Lava PLAYBOOK section 3). download-proxy (Python): httpx/requests honor HTTP_PROXY/HTTPS_PROXY/ALL_PROXY/NO_PROXY env natively — add an explicit BOBA_UPSTREAM_PROXY config that sets these for tracker-bound clients, with loopback bypass (NO_PROXY=127.0.0.1,localhost,jackett). qBitTorrent-go: set http.Transport.Proxy (socks5 native, remote DNS) from a BOBA_UPSTREAM_PROXY env — mirror Lava internal/httpx/proxy.go. Jackett: has a built-in proxy setting (configure via its API/ServerConfig). Deploy gotcha (port): the env must be FORWARDED into the containers (docker-compose.yml env / the

boba-ctl deploy) — Lava bug was a missing allow-list entry. Verify on distroless via podman inspect, not exec printenv. TDD: a test with a local proxy asserts the service tracker request traverses it; falsifiability: disable the wiring so the test fails. Source: docs/PORTING-FROM-LAVA.md. Per audit RD2-15 [P0]: Create tracked workable items (BOB-064..067) for the four Lava-porting findings, citing implementing commits as evidence, closed as Implemented.

BOB-068 — RD2-00: unattributed, unreviewed Auto-commit mechanism pushing to main

Status: In progress **Type:** Bug **Severity:** High

RD2-00: unattributed, unreviewed Auto-commit mechanism pushing to main

[BOB-136 adoption audit 2026-08-21 -> In progress] Work landed but acceptance NOT met. a7e55f9 completed the Phase-1 investigation (no daemon; shared-checkout race confirmed) and 0972cbc added commit-push-all.sh -scope + challenge, described in its own message as an 'Interim tooling remedy for BOB-068 while the §11.4.179 architectural fix (task #67 proposal) is designed'. 35e53db is a DRAFT PROPOSAL only. The §11.4.179 clone-isolation fix is not landed, so the item stays open.

BOB-069 — RD2-40: §11.4.238 QA-discovery-channel ledger — ongoing coverage-escape backfill

Status: In progress **Type:** Task **Severity:** High

RD2-40: §11.4.238 QA-discovery-channel ledger — ongoing coverage-escape backfill

[BOB-136 adoption audit 2026-08-21 -> In progress] a4173c8 stood up the ledger + 4 followups; 709b06c closed 2 Important review findings (CODEGRAPH-1.5.0-GITIGNORE entry, BOB-108 filed). The BOB-009/010 evidence_path gap this item names as remaining open was closed under BOB-086 (f78f383). NOT terminal by the item's own text: it is explicitly scoped 'P1-ongoing', with the full retroactive audit of the ~50 GA-NN/RD2-NN findings deliberately NOT attempted.

BOB-074 — RD2-07: DDoS-class testing fully absent from the mandated test-type matrix

Status: In progress **Type:** Task **Severity:** Medium

RD2-07: DDoS-class testing fully absent from the mandated test-type matrix

[BOB-136 adoption audit 2026-08-21 -> In progress] ae2b5cb authored challenges/scripts/ddos_resilience_challenge.sh (424 lines, 3 public surfaces) + docs/testing/{ddos_resilience,test_type_matrix}.md, so DDoS-class testing is no longer 'fully absent'. But of the three coverages this item's Fix line enumerates, a grep of the shipped challenge finds flood=2 hits, malformed=0, exhaust=0 — malformed-request-flood and resource-exhaustion-under-attack are not covered. 258b7db filed the 6 followups (BOB-109..114). Partial, not complete.

BOB-077 — RD2-10: Identify second host running the Auto-commit rsync/sync mechanism (OPERATOR-DECISION)

Status: Operator-blocked **Type:** Task **Operator-Block-Details:** WHAT: OPERATOR must identify which second host holds push credentials + confirms whether the rsync/sync mechanism is intentional or a stale job to retire. This session cannot inspect a host it has no access to (§11.4.6/§11.4.101). WHY: This session cannot inspect or reach the +0500 host that produced the Auto-commit fast-forwards (§11.4.6 no-guessing, §11.4.101 reversible-safe default). UNBLOCK: [A] operator names the second host + confirms intentional (proceed to RD2-11 wiring) · [B] operator confirms stale/misconfigured (retire the job) · [C] operator confirms the second live Claude session/device is the source (Rev-6 Update 2 root cause, proceed to per-session discipline enforcement) WHO: Operator **Severity:** High

[Backfill from GOVERNANCE_AUDIT_2026-08-08_ROUND2.md RD2-10, P0 OPERATOR-DECISION] Root-caused as far as this host allows (see RD2-00 Update): reflog proves the two newest commits arrived via plain pull: Fast-forward from a +0500 host, matching this project established 2026-06-28 cross-host rsync-sync pattern (cdb555f/55b8671). Needs operator input to go further — which second host runs it, and is it the intended mechanism (just needs a real message + review gate) or a stale job to retire. Not auto-executed (§11.4.6/§11.4.101 — this session cannot inspect or guess at a host it has no access to). Rev-6

update: root cause narrowed to a second live Claude session (Opus 5, same +0500 host) that independently landed constitution 177f2b0. Priority: P0 (operator-blocked).

BOB-078 — RD2-11: Once identified, wire Auto-commit mechanism through §11.4.234 dedicated commit/push script OR retire it

Status: Queued **Type:** Task **Severity:** High

[Backfill from GOVERNANCE_AUDIT_2026-08-08_ROUND2.md RD2-11, P0] Once identified (RD2-10): either wire it through a real §11.4.234 dedicated commit/push script (with the hook-validation stages this project already has via .claude/settings.json PreToolUse guard, extended to a real pre-commit content check) or shut it down if it serves no purpose. Priority: P0. Blocked on RD2-10.

BOB-079 — RD2-12: Retroactive attributed history notes for GA-18/21/22/25/26/27 changes (never rewrite published history)

Status: Queued **Type:** Task **Severity:** Medium

[Backfill from GOVERNANCE_AUDIT_2026-08-08_ROUND2.md RD2-12, P1] Retroactively give proper, attributed, reviewed commit messages / history notes to the technically-correct changes the Auto-commit mechanism already made (GA-18, GA-21, GA-22, GA-25, GA-26, GA-27) — this can be a documentation/changelog note rather than a history rewrite (§11.4.113 — never rewrite published history), citing the git-history evidence per §11.4.124 investigate-before-attribute pattern. Priority: P1.

BOB-080 — RD2-13: Retroactive Fable-xhigh code review of the substantive Auto-commit diffs

Status: Queued **Type:** Task **Severity:** High

[Backfill from GOVERNANCE_AUDIT_2026-08-08_ROUND2.md RD2-13, P1] Retroactively run the mandatory independent Fable-xhigh code review (§11.4.125/§11.4.142/§11.4.209) against the substantive diffs the Auto-commit mechanism introduced (start.sh three new functions especially, since they have zero test coverage — see Root Cause 4 / RD2-24) — even though the changes already landed, a post-hoc review closes the governance gap and will surface RD2-24 (GA-27 missing tests) if not already caught. Priority: P1.

BOB-082 — RD2-15: Create BOB-064..067 workable items for the four Lava-porting findings (closes GA-05)

Status: Queued **Type:** Task **Severity:** High

[Backfill from GOVERNANCE_AUDIT_2026-08-08_ROUND2.md RD2-15, P0 — closes GA-05] Create tracked workable items (BOB-064..067, via the workable-items tool — never raw MD edits) for the four Lava-porting findings, citing implementing commits as evidence, closed as Implemented. GA-05 evidence: `grep -in lava|BOB-06[4-7] docs/Issues.md docs/Fixed.md` returns zero hits. Priority: P0. NOTE: BOB-064..067 have been created 2026-08-10 as Task/Queued (the four Lava P1..P4 items); the closed-as-Implemented status flip (citing per-finding implementing commits) remains owed under this item.

BOB-085 — RD2-18: Create top-level Boba proxy/merge-service v1.0.0 readiness ledger (closes GA-10)

Status: Queued **Type:** Task **Severity:** Medium

[Backfill from GOVERNANCE_AUDIT_2026-08-08_ROUND2.md RD2-18, P2 — closes GA-10] Create the top-level Boba (proxy/merge-service) v1.0.0 readiness ledger (GA-10) — dedupe with the browser_extension existing one as the template. GA-10 evidence: only docs/RELEASE_READINESS_20260616.html/.md/.pdf (dated point-in-time snapshot) and the extension own ledger exist; no top-level proxy/merge-service ledger created. Priority: P2.

BOB-087 — RD2-20: Wire docs_chain / commit-seam sync hook per §11.4.106(F) so DB writes cannot land without MD mirror

Status: Completed (→ Fixed.md) **Type:** Task **Severity:** High

[Backfill from GOVERNANCE_AUDIT_2026-08-08_ROUND2.md RD2-20, P0] Wire (or fix) the docs_chain / commit-seam sync hook per §11.4.106(F) so a docs/workable_items.db write can never again land without its MD mirror in the same commit — this is the mechanical fix that prevents Root Cause 2 from recurring, not just a one-time catch-up. Priority: P0.

Progress 2026-08-21: MEASURED against all three seams §11.4.106(F) names, not one. COMMIT seam: COVERED — scripts/hooks/docs-sync-commit-seam.sh is invoked from scripts/commit-push-all.sh at BOTH commit call sites (the -scope branch and the git add -A branch) via _docs_sync_seam_check, after staging and before git commit, exiting 1 on refusal; there is no third path to git commit in that script. Proven in BOTH directions on temp copies with the real DB sha256 unchanged: an MD-side body edit is detected and named (self-test golden-bad, victim BOB-008), and a real engine DB write with the Markdown deliberately left stale is detected and named (golden-bad, BOB-084) — that second direction is the one the item's own text claims — while a clean tree stays silent (negative control, no §11.4.201(1) false positive). BUILD seam: COVERED — pre_build_verification.sh invariant 17 runs validate AND diff with -issues/-fixed passed explicitly (never the flagless form BOB-155 fixed), invariants 18/22 cover the export leg with a real-invocation assertion, invariant 24 runs the real docs_chain engine verify -all; RESIDUAL: no CHECK 3 equivalent there, so the build seam inherits diff's blindness to the body_md class (the BOB-136 class). CONSTITUTION-PULL seam: NOT COVERED — a grep for workable-items|docs-sync|docs_chain|11.4.106 returns 0 in BOTH constitution/scripts/post_update_hook.sh and scripts/verify-all-constitution-rules.sh, control-needled so the zeros are sight not blindness (needle 'skill' 38 hits, 'covenant_propagation_suite' 7 hits, negative control 0); of the 172 gates under constitution/scripts/gates/ only two mention the engines and both are anchor-literal presence gates that compare no DB against any Markdown, and config/constitution-sweep.conf adds no such check. So a constitution pull can be treated as canonical with the tracker never re-compared. REMAINS: wire the already-existing seam into scripts/verify-all-constitution-rules.sh BY REFERENCE (bash scripts/hooks/docs-sync-commit-seam.sh -files docs/Issues.md docs/Fixed.md docs/workable_items.db), reporting PASS/FAIL/SKIP-with-reason in the sweep's own vocabulary and never a silent pass on an absent tool — a wiring change, not a second

implementation (§11.4.227). NOT DONE this round: that file sits outside the working brief's declared file scope, so the gap is named and the item stays open rather than the scope being exceeded (§11.4.6). EVIDENCE. docs/qa/BOB-087/seam-coverage-measurement.md.

BOB-088 — RD2-21: Complete/verify README Tracked-Items + Status Documents table row-completeness (GA-07 remainder)

Status: Queued **Type:** Task **Severity:** Medium

[Backfill from GOVERNANCE_AUDIT_2026-08-08_ROUND2.md RD2-21, P1] Complete/verify the README Tracked-Items + Status Documents table row-completeness (GA-07 remaining half). Not independently re-verified this round whether every mandated doc (CONTINUATION.md, Issues.md, Fixed.md, PORTING-FROM-LAVA.md, both new GA/RD2 audit docs, every Status.md/Status_Summary.md pair) actually has a row. Priority: P1.

BOB-090 — RD2-25: HelixQA Challenge entry exercising all three start.sh subcommands end-to-end against real compose stack

Status: In progress **Type:** Task **Severity:** Medium

RD2-25: HelixQA Challenge entry exercising all three start.sh subcommands end-to-end against real compose stack

[BOB-136 adoption audit 2026-08-21 -> In progress] a9366c9 bumped submodules/helixqa to HelixDevelopment/qa@00c5ca4 adding banks/boba-start-sh-reload.yaml (3 cases, all three subcommands). The bank EXISTS but has never been EXECUTED: the commit states honestly that the helixqa binary was not built in-session (§11.4.3 SKIP feature disabled by config) and 'live bin/helixqa list demo is deferred', with only YAML-parse evidence captured. This item's acceptance is end-to-end execution against the real compose stack — artifact-class evidence cannot close a runtime-class acceptance (§11.4.226).

BOB-092 — RD2-27: Remove test_live_stack_evidence.py:265 nnmclub SKIP-on-404 fallback + verify live 200 (closes GA-13)

Status: Queued **Type:** Bug **Severity:** Medium

[Backfill from GOVERNANCE_AUDIT_2026-08-08_ROUND2.md RD2-27, P1 — closes GA-13] tests/e2e/test_live_stack_evidence.py:265 — with the live stack up, confirm curl :7187/api/v1/auth/nnmclub/status returns 200 (redeploy first if source≠artifact drift persists — see Root Cause 6), remove the skip fallback entirely, let the real assertion run. Priority: P1.

BOB-093 — RD2-28: Live compose bring-up + verify rutracker ReDoS regex bounds deployed to container (closes runtime half of GA-12)

Status: In progress **Type:** Task **Severity:** High

RD2-28: Live compose bring-up + verify rutracker ReDoS regex bounds deployed to container (closes runtime half of GA-12)

[BOB-136 adoption audit 2026-08-21 -> In progress] 1c0389a proved 3 of the 4 sub-steps with strong runtime evidence: the bounded regex is confirmed in the DEPLOYED /config/qBittorrent/nova3/engines/rutracker.py inside the running container (sha256 76a6bd2e..), with a self-validated challenge + \$11.4.115 RED (mutated unsafe form FAILs) and GREEN verdict JSON. The 4th sub-step is NOT met: 'capture timing of a large rutracker result page (<2s)' was not exercised — in docs/qa/BOB-093/live_search_smoke.txt rutracker returned status=empty, results_count=0 in 164ms, so no large page was ever timed.

BOB-094 — RD2-29: Author tests/stress/test_tracker_fetch_stress_chaos.py with \$11.4.85 fault injection

Status: In progress **Type:** Task **Severity:** Medium

RD2-29: Author tests/stress/test_tracker_fetch_stress_chaos.py with §11.4.85 fault injection

[BOB-136 adoption audit 2026-08-21 -> In progress] The suite landed (test file at 4b7b21d, evidence + RED-on-mutated-classifier capture at f3ca131) with 9 test functions: 2 stress, 3 boundary, 3 chaos (network_drop, midflight_kill, input_corruption), 1 category_map. This item explicitly requires fault injection across process-kill, network-fault AND resource-exhaustion. Only 2 of those 3 classes are present — there is no resource-exhaustion scenario in tests/stress/test_tracker_fetch_stress_chaos.py. Partial.

BOB-095 — RD2-30: Author tests/stress/test_scheduler_hooks_sse_stress_chaos.py for Go-side triangle

Status: In progress **Type:** Task **Severity:** Medium

RD2-30: Author tests/stress/test_scheduler_hooks_sse_stress_chaos.py for Go-side triangle

[BOB-136 adoption audit 2026-08-21 -> In progress] d5c8c3e authored qBitTorrent-go/internal/service/sse_broker_stress_chaos_test.go (462 lines, 6 scenarios, 2 RED captures) and found+fixed a real double-unsubscribe 'close of closed channel' defect. But this item names a THREE-component triangle: sse_broker.go + internal/api/hooks.go + internal/api/scheduler_api.go. Only sse_broker.go is covered — there is no stress_chaos test anywhere under qBitTorrent-go/internal/api/. 1 of 3, partial.

BOB-097 — RD2-32: Author DDoS-class coverage for exposed download-proxy/merge endpoints (canonical impl of RD2-07)

Status: Queued **Type:** Task **Severity:** Low

[Backfill from GOVERNANCE_AUDIT_2026-08-08_ROUND2.md RD2-32, P3] Author DDoS-class coverage (RD2-07) for the exposed download-proxy/merge endpoints. Priority: P3.

BOB-100 — RD2-39: Bump submodules/jackett one commit (canonical impl of RD2-09)

Status: Queued **Type:** Task **Severity:** Low

[Backfill from GOVERNANCE_AUDIT_2026-08-08_ROUND2.md RD2-39, P3] Bump submodules/jackett one commit (RD2-09).
Priority: P3.

BOB-101 — GA-19/RW-09: Is -profile go parity still a release goal? (gates RW-10..13) — OPERATOR-DECISION

Status: Operator-blocked **Type:** Task **Operator-Block-Details:**
WHAT: OPERATOR must decide whether Go -profile go parity remains a v1.0.0 release goal; gates downstream RW-10..13. WHY: §11.4.66 forbids autonomous decision on release scope; §11.4.122 forbids silent removal of an existing capability. UNBLOCK: [A] operator states YES — parity remains a release goal (schedule RW-10..13 work) · [B] operator states NO — mark parity Obsolete/superseded per §11.4.90 (reason=feature-removed with operator citation, §11.4.122) WHO: Operator **Severity:** High

[Backfill from GOVERNANCE_AUDIT_2026-08-08_ROUND2.md GA-19/RW-09, OPERATOR-DECISION] Confirmed unchanged: zero time.Ticker anywhere in qBitTorrent-go, no enricher package, zero exec.Command fan-out. Is -profile go parity still a release goal? Gates RW-10..13. Surfaced only, not auto-executed, per §11.4.66. Priority: OPERATOR-DECISION.

BOB-102 — RW-05: LAN-exposure threat model — bind tunnel 127.0.0.1 or keep 0.0.0.0? — OPERATOR-DECISION

Status: Operator-blocked **Type:** Task **Operator-Block-Details:**
WHAT: OPERATOR must decide LAN-exposure posture: bind tunnel to 127.0.0.1 (loopback-only) OR keep 0.0.0.0 (LAN-reachable, relying on post-RD2-22 complete auth coverage). WHY: This is a security-posture policy call the agent cannot make autonomously (§11.4.66 operator-decision, §11.4.101 high-blast-radius reversible-only-if-explicit). UNBLOCK: [A] operator states BIND 127.0.0.1 (loopback-only; agent will change compose/service binding + regression-test LAN unreachability) · [B]

operator states KEEP 0.0.0.0 (rely on §RD2-22 completed auth coverage; agent will add a permanent §11.4.135 LAN-auth regression guard) WHO: Operator **Severity:** Medium

[Backfill from GOVERNANCE_AUDIT_2026-08-08_ROUND2.md RW-05 ungrouped, OPERATOR-DECISION] LAN-exposure threat model — bind the tunnel 127.0.0.1 or keep 0.0.0.0 with the now-complete auth coverage (post Root Cause 3)? Priority: OPERATOR-DECISION.

BOB-104 — §11.4.238 followup: CodeGraph 1.5.0 nested-.gitignore regression challenge

Status: In progress **Type:** Task **Severity:** Medium **Created-By:** Claude

Coverage-escape followup (docs/QA_DISCOVERY_LEDGER.md, BOB-075 agent-code-reading finding, commit e6162f7): CodeGraph 1.5.0 (up from documented 0.9.9) walked into nested-.gitignore-excluded frontend/node_modules and extension/node_modules (32,260 files / 514,456 nodes vs the 2026-06-06 baseline of 509 files / 8,906 nodes) instead of honoring frontend/.gitignore + extension/.gitignore per git check-ignore -v. Author a challenge (challenges/scripts/codegraph_gitignore_honor_challenge.sh or equivalent, e.g. wrapping 'codegraph doctor -sniff-gitignore-honor' if that subcommand exists, else a real re-index + file-count assertion) with §11.4.115 RED_MODE polarity: RED_MODE=1 reproduces the blowup against the live nested-.gitignore tree, RED_MODE=0 asserts the resync stays within the documented baseline order of magnitude. Full evidence: docs/codegraph/Status.md lines 91-118. UPSTREAM FILED 2026-08-18: real upstream repo is github.com/colbymchenry/codegraph (npm package @colbymchenry/codegraph, confirmed via package.json + gh repo view; NOT vasic-digital/codegraph, correcting this ledger's original assumption). Issue: <https://github.com/colbymchenry/codegraph/issues/1567> (full reproduction evidence: git check-ignore -v re-verified first-hand; two bounded synthetic repro attempts up to 63 nested .gitignore files / 960 files against the actual installed v1.5.0 binary did NOT reproduce the blowup in isolation - honest negative result, root cause not pinned to a specific line in either scanning implementation). Draft PR (diagnostic only, not a behavioral fix): <https://github.com/colbymchenry/codegraph/pull/1568> - adds a logDebug() call so a future report can confirm which of the two independent gitignore-respecting code paths (git-ls-files-based vs filesystem-walk fallback) actually ran. Both PRs/issue filed via SSH (Hard Stop #2) from a fork at github.com/milos85vasic/codegraph. Status: In-progress pending upstream

maintainer triage/merge - boba-side closure of this item still needs the RED/GREEN challenge script (§11.4.115) authored per the original scope, independent of upstream's response.

BOB-106 — §11.4.238 followup: §11.4.84 quiescence-check helper for the unattributed auto-commit path

Status: Queued **Type:** Task **Severity:** Medium **Created-By:** Claude

Coverage-escape followup (docs/QA_DISCOVERY_LEDGER.md RD2-00/BOB-068 entry, docs/GOVERNANCE_AUDIT_2026-08-08_ROUND2.md RD2-00): 20 bare 'Auto-commit' commits exist in this repo's git history (confirmed via `git log -oneline -all -grep`, e.g. 54e313f/9c8f684/743097a/de9270b/1c36777/41179c2/7c529ca) with no ATM-NNN reference and no TDD trail, landing via ordinary git pull fast-forward from a second session/host with push access to the same remotes (mismatched commit timezone vs the investigating host, per RD2-00 Update). §11.4.84 working-tree-quiescence has no mechanical guard on this path — no gate flags a commit reaching main with a bare/templated message and no ticket citation. Author a §11.4.84 quiescence-check helper (e.g. `challenges/scripts/no_unattributed_autocommit_challenge.sh`) that scans the commit range since the last known-good release tag and FAILs on any commit message matching a closed bare/templated pattern (e.g. `^Auto-commit$, ^sync:`) with no ATM-NNN or task/PR reference, wired into `scripts/pre_build_verification.sh` or the §11.4.234 `commit-push-all.sh` entrypoint. BOB-068 (RD2-00) remains the tracking item for identifying/stopping the source; this item is specifically the new automated CHECK.

BOB-107 — §11.4.238 followup: pre- dispatch existence check for subagent task-brief source inputs

Status: Queued **Type:** Task **Severity:** Medium **Created-By:** Claude

Coverage-escape followup (docs/QA_DISCOVERY_LEDGER.md SCRATCH-LOSS-2026-08-18 entry, evidence: `.superpowers/sdd/task-phase1a-report.md` line 21): the Phase 1a subagent (a1cc331d) discovered at task start that 5 source files its brief named as required reads (`curriculum_amendment_plan_v1.md`, `ai_curriculum_modules_27_35_extracted.md`, and 3

curriculum_analysis_modules_*.md gap analyses) were absent from the session scratchpad — root cause per that subagent's own investigation: the prior producer subagent (ae59171f) hit its session rate limit (a §11.4.147(e) API-quota crash) before writing them. curriculum_amendment_plan_v1.md remains absent from the live scratchpad as of this session's re-check. No mechanical check verifies a task brief's declared input files exist and are non-empty before the downstream consumer subagent is dispatched. Author a pre-dispatch precondition helper (project-side orchestration tooling, e.g. a small script the conductor runs before Task/Agent dispatch when a brief names required input paths) that fails closed with an actionable 'missing input, respawn the producer' message rather than silently letting a downstream agent proceed on absent evidence and fabricate content unsupported by its named sources.

BOB-109 — BOB-074 followup: scaling-class test coverage absent from mandated test-type matrix

Status: Queued **Type:** Task **Severity:** Medium **Created-By:** Claude

docs/testing/test_type_matrix.md's §11.4.27 test-type audit found zero scaling-class coverage anywhere in the tree (no scaling-tagged directory, test file, or HelixQA bank distinguishes growing-dataset/tracker-count/concurrent-user scale-out from stress-under-burst). Scope at least one scaling dimension, e.g. tracker-count scale-out in merge search against challenges/helixqa-banks/boba-services.yaml's tracker set, or the qbittorrent-proxy-go -profile go swap, with a real measured baseline.

BOB-110 — BOB-074 followup: UX-class test coverage (accessibility/usability) absent

Status: Queued **Type:** Task **Severity:** Medium **Created-By:** Claude

docs/testing/test_type_matrix.md's §11.4.27 test-type audit found UI functional coverage (Vitest + Playwright) but nothing framed around usability/accessibility/UX outcomes specifically. Scope an axe-core or equivalent accessibility pass over the Angular frontend, covering WCAG checks, keyboard-nav coverage, and screen-reader labeling.

BOB-111 — BOB-074 followup: configure real rate limiting for boba's 3 public HTTP endpoints

Status: In progress **Type:** Task **Severity:** High **Created-By:** Claude

Source inspection across the whole stack (2026-08-18) verified no rate-limit mechanism exists for :7185 (qBittorrent WebUI proxy), :7187 (merge search service), or :7189 (boba-jackett) - no slowapi/limiter/throttle import in download-proxy/src/, no rate-limit middleware in qBitTorrent-go/internal/middleware/ (only cors.go + logging.go) or internal/jackettapi/ (only auth/cors middleware tests, no rate middleware), and no nginx/reverse-proxy service in docker-compose.yml. Candidate remediations documented in docs/testing/ddos_resilience.md: nginx-in-container reverse proxy with limit_req_zone (most portable, adds a container); a FastAPI slowapi dependency for the merge-search service; a Gin rate-limit middleware for boba-jackett following the existing internal/middleware/ pattern. qBittorrent's own WebUI bandwidth-shaping settings were NOT verified to cover request-rate (only bandwidth) - do not assume they close this gap without checking.

BOB-114 — BOB-074 followup: self-validation golden-bad fixture for the rate-limit detector

Status: Queued **Type:** Task **Severity:** Medium **Created-By:** Claude

challenges/scripts/ddos_resilience_challenge.sh's -self-validate mode currently only ships a golden-bad fixture for the crash-resistance detector (per §11.4.107(10)/§11.4.201). The rate-limiting detector has no matching golden-bad fixture proving it would actually FAIL a synthetic no-rate-limit-enforced artifact, so an unvalidated rate-limit detector could silently pass a broken/absent rate-limit deployment. Add a synthetic fixture (e.g. a stub server that never returns 429/503 under burst) and assert the detector correctly reports the absence, closing the self-validation gap for this detector class.

BOB-120 — 3rd forced-logout incident 2026-08-18 23:45:49 — SIGKILL user@1000 + preventive-timer-inside- user-slice architectural gap

Status: Queued **Type:** Bug **Severity:** Critical **Created-By:** AI

3rd forced-logout incident 2026-08-18 23:45:49 — SIGKILL
user@1000 + preventive-timer-inside-user-slice architectural gap

[BOB-136 adoption audit 2026-08-21 -> DELIBERATELY LEFT QUEUED] The CM-CLOSURE-SEAM-BINDS gate classifies this row as a CONTRADICTION 'via closes, commit d84d226'. That is a CARRIER match, not a closure. The matching text in d84d226 is 'closing BOB-120 requires an out-of-user-scope watchdog, not more documentation' — a statement of what closure would take, and the same commit states 'New architectural finding, filed as BOB-120 (Critical, left Queued)'. d84d226 FILED this item; it did not close it. The architectural fix remains unlanded: BOB-121, which carries it, is still Ready for testing and its own body states it is NOT CLOSED, with two operator decisions owed and the watchdog UNTESTED AGAINST A REAL FORCED LOGOUT (survival inferred from cgroup topology, not observed). Moving this row on documentation alone would be precisely the §11.4.238 coverage-escape bluff this item's own text warns against. Status unchanged; the gate finding is expected to persist until the out-of-scope watchdog actually lands and is verified.

BOB-121 — External watchdog for the forced-logout architectural gap (task #85, incident #3)

Status: Ready for testing **Type:** Task **Severity:** Important
Created-By: Claude

Phase 1 design-only proposal: the BOB-116/task-77 resource-pressure preventive systemd -user timer runs inside user@1000.service, the exact pool it monitors, so it cannot fire when that pool dies (proven by incident #3, docs/qa/BOB-120/, 22:42+22:57 fires then blocked 23:45:49-23:49:00). Recommends Option B (user crontab reusing pre-existing crond.service in system.slice, no new root service) kept alongside the existing timer, with Option A (new root-owned systemd unit) as escalation path if Phase 1.5 live cron/cgroup verification is adverse. Proposal: docs/proposals/external-watchdog-for-forced-logout-architectural-gap.md. Operator decision required per §11.4.66 before any implementation - NOT implemented in this task.

Progress 2026-08-21: Flight recorder built, installed, and recording. First finding: NOTHING was watching at all — the root watchdog is LoadState=not-found and the BOB-116 user timer inactive. The blocking spike REFUTED the proposal's rationale while confirming its conclusion: this vixie-cron does invoke pam_systemd so the tick lands in session-N.scope, but that scope is a SIBLING of user@1000.service, and the real incident killed exactly one unit (73 'user@1000.service: Killing process' lines while session-18.scope merely deactivated). So the recorder does not need to survive — its SCHEDULER does, and crond in system.slice survived all seven incidents. Captures the pre-event memory/PSI/thread ramp that nothing post-hoc can recover, plus boot id, the gap itself, unit start timestamp, cgroup pids, and OOM cgroup attribution. Evidence: 5 records under real cron at exact 60s spacing; a staged analogue on a DISPOSABLE unit reaching 'VERDICT: SESSION TEARDOWN'; golden-good against the REAL BOB-120 log giving the correct diagnosis (k_unit_kill=1, k_oom=0); healthy-host quiet under load 8.15. NOT CLOSED — two operator decisions are owed: whether to keep it installed (one crontab line, zero signals, zero power verbs, 32K, reversible via uninstall.sh), and the root system.slice watchdog which is written but needs one su. UNTESTED AGAINST A REAL FORCED LOGOUT: survival is inferred from cgroup topology, not observed, and it does NOT hold against a whole-slice sweep or KillUserProcesses=yes. ## BOB-129 — Potential production slowapi/starlette defect flagged by Task 105 subagent

Status: Fixed (→ Fixed.md) **Type:** Bug **Severity:** Medium

Task #105 subagent (fixing 9 slowapi test failures) reported honestly that the same slowapi/starlette incompatibility likely hits the production /search and /search/stream endpoints under real HTTP traffic — evidence: the FastAPI TestClient (which goes through the full ASGI middleware stack like real requests do) reproduces the same isinstance() failure pattern the 9 test failures exhibited. Not yet reproduced against the running boba stack because the qbittorrent-proxy container currently exposes no host ports (running on gluetun network stack). Recommended investigation: (1) confirm defect by triggering /search kickoff through gluetun network stack, (2) if reproduced, determine whether the fix belongs in production code (adding response: Response params) or a version pin (slowapi vs starlette compat) or a middleware refactor. §11.4.238 discovery-channel escape prevention: manual QA must NOT be the discoverer. §11.4.108 Layer 3 verification: needed on a clean deployment before any release.

BOB-131 — qbittorrent-proxy podman common crash — pre-existing, surfaced during BOB-129 investigation

Status: Fixed (→ Fixed.md) **Type:** Bug **Severity:** Medium

BOB-129 subagent found qbittorrent-proxy container DEAD mid-investigation (podman common crash). Recovery via `./start.sh -no-build` worked. Pre-existing, unrelated to BOB-129 slowapi work but surfaced by it. Investigation needed: (a) how long was container dead before discovery? (b) what triggered the common crash? (c) is there a §11.4.144 always-follow / §11.4.128 always-record signal we should add to detect this class earlier? Post-recovery: container now unhealthy (see BOB-132). §11.4.238 discovery-channel escape: was originally found by a subagent investigating something else, not by dedicated container health monitoring.

Closed 2026-08-21 — THE PREMISE OF THIS TICKET IS FALSE, and that is the finding.

No common process crashed. This item conflated TWO UNRELATED EVENTS. Over the full 7.5-day journal retention the only common messages above warn are common REPORTING failures (Failed to create container: exit status 1, Failed to write 137 to exit file) — never crashing. A common crash would put common in a kernel segfault line; the only such line in a week is a python3 one.

EVENT 1, once, 2026-08-20 17:56:26 CEST: python3[314359]: segfault at 70 ... in libpython3.12.so.1.0, with the container's own stdout ending mid-" File " — the dumper died writing it. Proven at machine level rather than inferred: the kernel's Code: bytes at IP were byte-matched against the library INSIDE the running container (MATCH), decoding to `mov r14,[r12]` (frame->f_executable = NULL) then `mov rax,[r14+0x70]` (code->co_filename) -> fault; the preceding `lea` resolves to the literal "File " with `edx=7`, its exact length and exactly the text the log truncated after. Self-healed in 0.03s via restart: unless-stopped. Zero recurrences since.

EVENT 2, the 14h06m43s absence: a HOST POWER-OFF (systemd-logind: The system will power off now!), container exited 0. restart: unless-stopped does not survive a power cycle, and boba-stack.service is linked but disabled.

ALL THREE LOOKALIKES EXCLUDED WITH EVIDENCE: cgroup OOM-kill (oom_kill 0, zero OOM lines in 7.5 days, flight recorder k_oom=0 across all 62 samples); cgroup memory-ceiling (REAL — memory.max=805306368 with 1584 memory.events in 48 min —

but that is reclaim, not a kill, and cannot produce SIGSEGV);
§12.12 thread exhaustion (ulimit -u 65536, peak 1559 = 2.4%, no
EAGAIN / 'failed to create new OS thread' anywhere).

THE ACTIONABLE RESIDUAL IS SPLIT OUT AS BOB-157 (High):
the crash vector is our OWN diagnostic — download-proxy/src/
main.py:135 calling `faulthandler.dump_traceback(all_threads=True)`
— hitting upstream python/cpython#116008 / #128400, fixed and
backported to 3.13/3.14 with NO 3.12 backport, on a container
shipping 3.12.13. Still armed.

DELIVERED: docs/guides/container-death-triage.md (a 6-class
decision table plus the three confusions above), docs/incidents/
2026-08-21-bob131-container-death-triage.md, and scripts/
diagnostics/bob131_container_death_triage.sh (TDD RED 7/8 ->
GREEN 8/8, deterministic 5/5, both §1.1 mutations FAIL —
collapsing oom_kill/ceiling breaks 5 fixtures including the negative
control). The investigating agent caught its OWN selftest passing
by race (an awk '...; exit' SIGPIPE) and fixed it before reporting.
BOB-135 — Test isolation: `test_list_hooks_after_create` fails in
bulk suite (Permission denied /config)

Status: Queued **Type:** Bug **Severity:** Low

Task #109 subagent found: tests/unit/
`test_merge_api_route_contracts.py::TestHooksEndpoint::test_list_`
`hooks_after_create` fails when run in bulk suite order with 'ERROR
`api.hooks:hooks.py:102 Failed to save hooks: [Errno 13]`
Permission denied: /config'. Passes in isolation (2.02s clean). Root
cause: full-suite ordering pollution — some earlier test leaves
state that makes hooks try to write to /config (which the test env
doesn't own). Pre-existing, unrelated to BOB-126/BOB-129 chain.
Fix strategy: identify the polluting test, add teardown or use a
proper tempdir fixture for hooks storage in the offending test.

BOB-136 — Closure seam does not bind: 4 tracker rows found stale in one sweep, and workable-items diff is blind to body_md drift

Status: Ready for testing **Type:** Task **Severity:** High

WHAT. A single 2026-08-20 sweep found FOUR workable-item
rows whose tracked state contradicted reality: BOB-076 (closed
by commit 99a486e on 2026-08-15, whose message literally says
"closes BOB-076", still Queued), BOB-117 (fix landed AND
covered by a permanent guard assertion, still Queued), BOB-091

(implemented under d1a8479 / a3b16bc / 2a0b543, still Queued), and BOB-008 whose Evidence field still quoted a diagnostic string that BOB-117 removed from source.

CORRECTION 2026-08-20 (11.4.6). An earlier revision of this item claimed the enabling defect was that “workable-items diff is blind to body_md drift”. That was WRONG and is corrected here rather than quietly edited. The engine is NOT blind. The FLAGLESS INVOCATION is:

```
workable-items diff --db docs/workable_items.db
-> "diff: DB and Markdown are in sync"      (opens ZERO .md
files)

workable-items diff --db docs/workable_items.db --issues docs/
Issues.md --fixed docs/Fixed.md
-> "~ BOB-008 body differs (md=1346 bytes db=1333 bytes)"  ->
"diff: 1 difference(s)"
```

Both measured on the same seeded body-only drift. An independent agent proved the mechanism with strace: the flagless form never opens any Markdown file. So this is a 11.4.201(6) FALSE-NUL of the CALLER, not a gap in the comparison logic. Importantly, pre_build_verification.sh invariant 17 already passes both flags, so the STANDING GATE was never blind — the drift above was not caused by a blind gate.

THE REAL DEFECTS, restated honestly. (1) The flagless form is a false-null trap: it reports a confident, reassuring “in sync” while checking nothing. Any caller using it is a green-reporting blind check. It should either default to the conventional Issues.md/ Fixed.md paths or REFUSE with a clear error, never return a clean verdict for a comparison it did not perform. (2) Work commonly lands under commit messages that do not carry the ATM id (GA-14/15/16 for BOB-091), so nothing mechanically links a commit back to its row, and closure depends on a human remembering. 11.4.227: an anchor done state is its SEAM landing, not its TEXT landing.

ACCEPTANCE. (a) The flagless diff invocation no longer returns a clean verdict without reading Markdown, with a paired 1.1 mutation proving it. (b) A repo-wide grep confirms no caller uses the flagless form. (c) A sweep that flags any non-terminal item whose id appears in a merged commit message, so done-but-open rows are found mechanically rather than by a human noticing. (d) Honest boundary: this does not claim to prevent all drift.

RELATED. scripts/hooks/docs-sync-commit-seam.sh (added under BOB-087) now enforces doc/DB sync at the commit seam and passes the flags explicitly, plus an independent body_md re-parse oracle.

EVIDENCE. docs/qa/BOB-117/closure-evidence.md, docs/qa/BOB-076/closure-evidence.md, docs/qa/BOB-091/closure-evidence.md.

Progress 2026-08-21: The closure seam now BINDS TO GIT. Finding: every pre-existing check compared the tracker to ITSELF (DB vs its own invariants, DB vs Markdown, Markdown vs exports) — none ever consulted git, so all four stayed green while a row said Queued and the fix had landed weeks earlier. The commit hook also EXITED EARLY when no tracker file was staged, skipping exactly the pure-code-fix commits whose rows fail to move. Built `check_cm_closure_seam_binds.sh` plus a `-message` commit-seam mode wired as `CHECK 5` before that early exit; it judges only the ids the PENDING message declares, so it is monotone and structurally cannot block on the backlog. Meta-test 29/29, live polarity 2 refused / 2 passed, §1.1 flips both golden-bads to MUTANT=0. PROVEN IN ANGER: it refused the very commit that introduced it, because that message declared BOB-136 and BOB-121 while both rows still read Queued. NOT YET CLOSED — 20 stale rows and 2 untracked ids remain to drain, and the gate is not wired into `pre_build_verification.sh` (that would break the build immediately on the pre-existing backlog; `-report-only` first). Honest boundary: it only sees work that DECLARES its id, so a fix landing under a message naming no item stays invisible — which is BOB-136's own second defect. ## BOB-137 — Merge service on 7187 wedges while the same process still serves 7186 (GIL starvation by one spinning thread)

Status: In progress **Type:** Bug **Severity:** High **Created-By:** Claude

Reported-Via: §11.4.202 reporting directive bug on 2026-08-20T14:46:52Z **Reported-By:** Claude

What (the report, verbatim): The merge service on port 7187 wedges after hours of uptime: the port stays bound and the container keeps reporting “healthy”, but every HTTP request hangs until the client times out. Measured 2026-08-20 16:41 UTC+2 on a container up 3h53m:

`curl http://localhost:7186/ -> HTTP 200 in 0.096s (proxy, same process)`
`curl http://localhost:7187/ -> HTTP 000 after 6.0s (merge service, WEDGED)`

Both ports are served by the SAME process (pid 2330069, fd 3 = 7186, fd 7 = 7187), so this is not a crashed worker - one loop inside a live process has stopped servicing requests while another in the same process is fine.

Thread state at the time of the wedge (/proc/2330069/task, 16 threads): - tid 2330529: state R, wchan 0 <- ONE thread spinning on CPU - tid 2330528: state S, wchan do_sys_poll <- the healthy 7186 poll loop - the other 14: state S, wchan futex_wait_queue

That is the GIL-starvation signature: a thread busy-looping in Python holds the GIL and every other thread queues on the GIL futex. 7186 survives because do_sys_poll releases the GIL; the 7187 async loop needs sustained GIL time to service a request and starves. Process CPU was 20.6% while idle.

Corroborating evidence: seven sockets held by the process sit in CLOSE-WAIT with unread bytes still in the receive queue (Recv-Q 162, 162, 162, 79, 1, 1, 1) - clients sent a request and hung up, and the app never read it or closed the fd. The listener had also accumulated an unaccepted backlog (Recv-Q 6 on the 7187 LISTEN socket) at first observation. The container log's last line is 2h before the observation, so the service processed nothing in that window.

NOT fd exhaustion: only 48 of 16384 fds were open.

ROOT CAUSE NOT ESTABLISHED. All four while True loops in the service (routes.py:166, search.py:1169, streaming.py:161, streaming.py:359) are correctly bounded and awaited, so the spin is not a naive unslept loop. py-spy could not attach to name the spinning frame - the host has kernel.yama.ptrace_scope=1, which denies non-child attach. Per the §11.4.102 Iron Law no fix is proposed until the spinning frame is identified.

Next diagnostic step: obtain a Python stack. Either run py-spy INSIDE the container (musl wheel), or set ptrace_scope=0 for the duration of one dump, or add a SIGQUIT/fault_handler.register() dump hook to main.py so the running service can be asked for its own stacks without ptrace.

DISCOVERY CHANNEL (§11.4.238): found by an agent probing the host by hand while investigating an unrelated test-suite result - NOT by automated QA. This is a coverage escape in its own right; see the sibling item on the health check that was structurally incapable of observing it.

Affected scope / file-scope manifest: download-proxy/src/main.py, download-proxy/src/merge_service/, download-proxy/src/api/streaming.py, docker-compose.yml (qbittorrent-proxy)

Reproduction / context: Leave the qbittorrent-proxy container running for several hours with search traffic (a full tests/security run is sufficient). Then: curl -max-time 6 http://localhost:7186/ returns 200; curl -max-time 6 http://localhost:7187/ returns 000.

Confirm with: `ss -ltnp | grep 7187 (unaccepted backlog) and awk '{print $3}' /proc//task/*/stat (one R thread, rest futex_wait_queue).`

Acceptance criteria: The spinning frame is identified from a real Python stack dump (not inferred), the busy-loop is fixed at its root, and a regression guard proves 7187 still answers after a sustained-traffic soak. Evidence: before/after curl timings on both ports plus a thread-state census showing no permanently-R thread.

[BOB-136 adoption audit 2026-08-21 -> In progress] 1dd7b0a ESTABLISHED the root cause with captured evidence (16 stack dumps showing Deduplicator.merge_results() called synchronously at search.py:914 on the event-loop thread; loop thread sustained 81-98% user-space CPU while all other threads showed `d_ utime=0`). The remediation landed under BOB-145 (0572b71, now Fixed), whose own text is headed 'BOB-145 - the 7187 wedge' and which refuted the assumed $O(N^2)$ cause by profiling. BOB-137 and BOB-145 therefore describe the SAME defect; per §11.4.214 that is a link/dedup decision, not a unilateral close, and no post-fix re-observation of the multi-hour wedge is recorded against BOB-137 itself. Left open pending that linkage decision.

Live verification 2026-08-21 — REDUCED BUT NOT ELIMINATED. Deliberately NOT closed.

The service was confirmed running POST-FIX bytes before any measurement, six independent ways: served sha256 == committed == worktree across 20 merge-service/api/main files; fix markers 10 inside the container vs 0 at the pre-fix commit; .pyc header decode showing embedded source mtime+size matching the actual file (so the import reused it and the loaded module was compiled from exactly this source); fix markers present in the LOADED bytecode; and the process starting 823s AFTER the fix hit disk. A first attempt at this comparison reported routes.py as stale — an INSTRUMENT ARTIFACT (marshal back-reference encoding differs between a fresh compile and a pyc load), caught before it became a false positive.

Soak, same script and concurrency, precondition reached (11,340 results over 516 tracker responses / 12 searches ~ 945 per merge, larger than BOB-145's N=800 maximum):

PRE-FIX (quoted from the recorded report): 22 of 26 probes dead on 7187 (84.6%)

POST-FIX (measured): 2 of 141 dead (1.4%)

The dead-count alone understates the residual: 25.5% of probes stalled >1s (<0.1s 65.2% / 0.1-1s 9.2% / 1-5s 18.4% / >=5s 5.7% / dead 1.4%). Both dead events show the exact BOB-137 asymmetry — 7186 answering in 0.077s while 7187 timed out at 10s — and

both coincide with the loop thread sampled at state R with wchan=0, the GIL-starvation signature (tid independently confirmed as the loop thread by its idle wchan do_epoll_wait).

WHY THIS ROW STAYS OPEN: the acceptance evidence this item names is '22/26 dead -> 0/N dead'. Achieved: 22/26 -> 2/141. The user-visible symptom — 7187 unresponsive while 7186 answers in the same process — STILL OCCURS, twice in 15 minutes. That is precisely BOB-145's own predicted residual: search.py:914 is still a plain synchronous call, and removing the symptom needs a change at that CALL SITE (offload or await), which BOB-145 explicitly scoped out.

Two criteria that ARE satisfied: no permanently-R thread (48 of 80 census samples had zero R threads), and the in-process 20s watchdog logged 0 stalls — with a control needle, since that same watchdog produced 167,971 bytes of dumps pre-fix. ## BOB-141 — CLAUDE.md claims the Go profile serves 7186/7187/7188 but its container binds only 7187 — doc contradicts the Dockerfile

Status: Queued **Type:** Task **Severity:** Low **Created-By:** Claude

Reported-Via: §11.4.202 reporting directive task on 2026-08-20T14:56:38Z **Reported-By:** Claude

What (the report, verbatim): CLAUDE.md's Architecture section states:

"qbittorrent-proxy-go (Go/Gin, opt-in via -profile go) - replaces the Python proxy on 7186, 7187, 7188"

The container cannot deliver that. Read from source rather than prose:

```
qBitTorrent-go/Dockerfile:16 CMD ["/app/qbittorrent-proxy"]
(ONE binary) qBitTorrent-go/Dockerfile:15 EXPOSE 7187 7188
(declares 2) cmd/qbittorrent-proxy/main.go r.Run(fmt.Sprintf("%d",
cfg.ServerPort)) (binds 1) internal/config/config.go:58
ServerPort = MERGE_SERVICE_PORT (default 7187)
```

So the qbittorrent-proxy-go container binds 7187 ONLY. webui-bridge is a SEPARATE binary (cmd/webui-bridge, /bridge/health, cfg.BridgePort) that this container never starts, and nothing in it binds 7186 either - although the compose service does set PROXY_PORT=7186 and BRIDGE_PORT=7188 in its environment, which reinforces the wrong impression. EXPOSE likewise declares a port nothing binds.

WHY THIS MATTERS BEYOND TIDINESS: this prose was used as the source of truth when first authoring config/served_ports.yaml, producing a manifest entry of [7186, 7187, 7188] for that service. The healthcheck gate built on it then FAILED a service whose healthcheck was already correct - a §11.4.201(1) false-positive

refusal caused directly by trusting the doc over the Dockerfile. The manifest was corrected against source; the doc was not, and will mislead the next reader the same way.

Either the doc is wrong, or the Go container is under-provisioned relative to intent (it should also run webui-bridge and a 7186 listener). Determining WHICH is part of this task – do not simply reword the doc to match the current binary if the intended design was a three-port container.

Related: the Go service's compose block sets PROXY_PORT and BRIDGE_PORT that no process in the container consumes, and EXPOSE lists 7188 unbound. Whichever way the above resolves, those should agree with reality afterwards.

Affected scope / file-scope manifest: CLAUDE.md (Architecture + Port Map), docker-compose.yml (qbittorrent-proxy-go env/EXPOSE), qBitTorrent-go/Dockerfile

Reproduction / context: Read qBitTorrent-go/Dockerfile:16 (single CMD) against CLAUDE.md's 'replaces the Python proxy on 7186, 7187, 7188'; confirm cfg.ServerPort resolves to MERGE_SERVICE_PORT=7187 in internal/config/config.go:58.

Acceptance criteria: Doc and container agree, with the direction of the fix decided deliberately (correct the doc, or provision the container to match the documented intent). PROXY_PORT/BRIDGE_PORT env and EXPOSE lines agree with what the container actually binds.

BOB-143 — Orphaned .worktrees/ dirs (46M, unresolvable gitdir) pollute gate scan scope and manufacture false BOB-126-class findings

Status: Queued **Type:** Bug **Severity:** Medium **Created-By:** Claude

Reported-Via: §11.4.202 reporting directive bug on 2026-08-20T15:08:18Z **Reported-By:** Claude

What (the report, verbatim): .worktrees/ci-split-workflows/ (13M) and .worktrees/completion-initiative-phase-0/ (33M) are ORPHANED: git worktree list reports only the main checkout, so neither is a registered worktree. Each contains a .git POINTER FILE whose target gitdir no longer exists, so git cannot resolve HEAD, branch, or status inside them – every query returns empty.

They are gitignored (.gitignore:122), so nothing tracks them and nothing will ever notice them drifting.

WHY THIS IS NOT COSMETIC: they pollute the scan scope of whole-tree gates and manufacture false findings. Measured 2026-08-20 by the §11.4.32 sweep:

- `cm_test_mock_pid_explicit_int` reported 2 violations at `tests/unit/merge_service/test_deadline_tunable.py:44` – BOTH inside these orphaned trees. The MAIN tree's copy of that file is already hardened (it sets `mock.pid = 12345` and patches `os.killpg/os.getpgid`) and passes the gate cleanly. The finding read as a live §11.4.263 / BOB-126-class defect and was not one.
- 6 of the 57 “missing anchor carrier” files flagged by the propagation gates were likewise `.worktrees/**`.

That is the §11.4.201(1) false-positive shape sourced from scan scope, and it costs real investigation time: a reader triaging “2 live BOB-126 violations” reasonably treats it as a host-safety emergency.

CLARIFICATION (established during triage, so the record is not alarming): these trees are NOT a kill(-1) vector. Their `download-proxy/src/merge_service/search.py` contains ZERO `os.killpg` calls – they predate that cleanup code entirely – so there is no unguarded signal call to reach. Additionally `pyproject.toml` sets `testpaths = ["tests"]`, so a plain `pytest` run does not collect from `.worktrees/`. No host-safety risk was found. The defect is scan-scope noise plus 46M of unreferenced disk.

WHY REMOVAL IS NOT DONE AUTONOMOUSLY (§11.4.122 / §11.4.124 / §11.4.101): because git cannot resolve their HEAD, it is NOT possible to prove their contents are merged into main. Deleting unprovable-provenance work is exactly the irreversible, operator-owned decision §11.4.122 reserves. Two options for the operator: (a) confirm removal (they are stale dev scratch dirs) – reversible only from backup, so a §9.2 pre-op backup should precede it; or (b) keep them and add `.worktrees/` to the gate scan-scope exclusion list as a §11.4.224(E)-fenced, checked-in, justified entry.

Either way the exclusion list is the cheaper immediate mitigation and does not destroy anything.

Affected scope / file-scope manifest: `.worktrees/ci-split-workflows/`, `.worktrees/completion-initiative-phase-0/`, gate scan-scope config

Reproduction / context: git worktree list shows only the main checkout; `git -C .worktrees/`

log -1 returns empty (gitdir target missing). Run the §11.4.32 sweep and observe `cm_test_mock_pid_explicit_int` report 2 violations, both under `.worktrees/`, while the main-tree file passes the same gate.

Acceptance criteria: Whole-tree gates no longer report findings sourced from orphaned worktrees: either the dirs are removed after operator confirmation with a §9.2 pre-op backup, or `.worktrees/` is added to a checked-in §11.4.224(E)-fenced exclusion list with justification. Verify by re-running the sweep and confirming zero `.worktrees`-sourced findings.

BOB-144 — /theme/stream calls the disconnect probe unguarded — fail-closed but via an uncaught traceback, inconsistent with the two SSE generators

Status: Fixed (→ Fixed.md) **Type:** Bug **Severity:** Low
Created-By: Claude

Reported-Via: §11.4.202 reporting directive bug on 2026-08-20T15:53:57Z **Reported-By:** Claude

What (the report, verbatim): `/theme/stream` in `download-proxy/src/api/routes.py:167` calls the disconnect probe with NO guard at all:

```
while True:
    if await request.is_disconnected():
        break
```

If that probe raises, the generator dies with an UNCAUGHT exception.

IMPORTANT — this is NOT the BOB-139 fail-open. The effect here is fail-CLOSED: the stream stops, and the enclosing finally: `store.unsubscribe(queue)` still runs, so the subscriber queue is released and nothing leaks. The outcome is CORRECT; the manner is not.

What is wrong with it: - it terminates via an uncaught traceback rather than a clean SSE close event, so the client sees a truncated stream instead of a reason; - the failure is not logged as a probe failure, so a systematically raising probe would show up as recurring tracebacks with no diagnosis; - it is inconsistent with the two SSE generators in

streaming.py, which after BOB-139 emit event: close with reason `disconnect_probe_failed` and log a warning. Three call sites of the same probe now behave two different ways.

The BOB-139 fix deliberately did not touch this file (ownership boundary, §11.4.119), and flagged it honestly rather than fixing it out of scope.

Acceptance: `/theme/stream` uses the same probe-failure discipline as the `streaming.py` generators — a clean close event with the `disconnect_probe_failed` reason plus a logged warning — proven in BOTH directions (§11.4.201(1)): a raising probe closes the stream cleanly AND a normally-connected client still streams uninterrupted. Prefer reusing the shared helper introduced by BOB-139 rather than a third copy of the logic (§11.4.251).

Honest boundary (§11.4.6): the production probe-failure rate is UNKNOWN — nobody has measured how often `is_disconnected()` actually raises. This is filed on the inconsistency and the missing diagnosis, not on a measured incident rate.

Affected scope / file-scope manifest: `download-proxy/src/api/routes.py` (~line 167, `stream_theme`)

Reproduction / context: Monkeypatch `Request.is_disconnected` to raise, open `/theme/stream`, observe the generator dies with an uncaught traceback rather than emitting event: close with reason `disconnect_probe_failed` as `streaming.py` now does.

Acceptance criteria: Same probe-failure discipline as `streaming.py` (clean close + `disconnect_probe_failed` reason + logged warning), verified both directions, reusing the BOB-139 helper rather than a third copy.

BOB-145 — Fix the 7187 wedge: offload and/or memoise `Deduplicator.merge_results` so $O(N^2)$ regex work stops blocking the asyncio event loop

Status: Fixed (→ `Fixed.md`) **Type:** Bug **Severity:** High
Created-By: Claude

Reported-Via: §11.4.202 reporting directive bug on 2026-08-20T16:01:43Z **Reported-By:** Claude

What (the report, verbatim): BOB-137 established the root cause: `Deduplicator.merge_results()` is called as a PLAIN SYNCHRONOUS CALL at `merge_service/search.py:914` and therefore executes ON the `asyncio` event-loop thread. While it runs, `uvicorn`'s only loop runs no callback — no accept, no read, no write — so port 7187 stops answering entirely. Measured episodes of 9m37s and 5m56s, self-clearing, recurring under ordinary traffic.

This item is the FIX. BOB-137 is the diagnosis and stays separate per the §11.4.102 Iron Law — the investigation deliberately applied no fix.

Three candidate directions, not yet chosen:

1. OFFLOAD — hand `merge_results` to a thread executor (`asyncio.to_thread / run_in_executor`). Smallest change, immediately unblocks the loop. Does NOT make the work cheaper, so a large enough merge still burns a core; and it introduces concurrency around `self._last_merged_results` and metadata, which must be checked for races before it is called safe.
2. MEMOISE — `_normalize_name` is called at FOUR sites per comparison, each running 5 re.sub, and `_extract_identity_from_result` twice more with a ~15-re.search chain. `lru_cache` has ZERO matches in that module today, so the seed's own normalisation is recomputed for every candidate. Caching the per-result normalisation is a large constant-factor win with no concurrency risk.
3. REDUCE THE COMPARISON SET — the $O(N^2)$ shape itself (blocking/bucketing by a cheap key before pairwise comparison). Largest win, largest change, highest risk of altering dedup RESULTS — which would need its own correctness evidence, not just a speed measurement.
4. then (a) is the likely order: (b) is risk-free and may alone bring the merge under the threshold, and (a) guarantees the loop is never blocked regardless.

MANDATORY for whoever takes this: - RED FIRST (§11.4.43/ §11.4.224): a test that FAILS on the current code by demonstrating the loop is blocked during a merge — e.g. assert a concurrent request to 7187 is served within a bounded time while a large merge runs. A pure speed benchmark is NOT the RED test; the defect is loop-blocking, not slowness. - BOTH POLARITIES (§11.4.201(1)): the loop stays responsive under a large merge AND dedup results are unchanged for the existing corpus. A fix that speeds up merging while changing which duplicates are detected is a different defect. - Use the existing instrument: scripts/

diagnostics/bob137_soak.sh reproduced the wedge 22/26; it is the natural GREEN check. - Honest boundary carried from BOB-137: measured growth is SUPER-LINEAR but not fully quadratic at $N \leq 400$ (2.4-3.4x per doubling vs 4.0). Worst case is quadratic BY CODE STRUCTURE. Do not cite “measured N^2 ”.

Affected scope / file-scope manifest: download-proxy/src/merge_service/search.py:914, download-proxy/src/merge_service/deduplicator.py

Reproduction / context: bash scripts/diagnostics/bob137_soak.sh — reproduced the wedge in 22/26 probes (7187 dead, 7186 alive throughout).

Acceptance criteria: Port 7187 answers within a bounded time while a large merge runs (loop never blocked), AND dedup results are unchanged for the existing corpus. Proven with the soak repro flipping from 22/26-dead to 0/N-dead, plus a dedup-equivalence check.

BOB-146 — Constitution §11.4.252 detector undercounts by 29% (30 vs 42 AST ground truth) — 4 distinct blind spots make its output a floor, not a census

Status: Fixed (→ Fixed.md) **Type:** Bug **Severity:** High
Created-By: Claude

Reported-Via: §11.4.202 reporting directive bug on 2026-08-20T16:10:50Z **Reported-By:** Claude

What (the report, verbatim): The constitution’s §11.4.252 detector (constitution/scripts/gates/cm_dangerous_combination_fail_closed.sh) UNDERCOUNTS by 29%. Measured against an independent AST instrument (structure, not text — a valid control needle per §11.4.201(7)) on boba’s plugins tree:

gate reports	: 30
AST ground truth	: 42
missed	: 12
false positives	: 0

A gate that misses 12 of 42 while reporting a confident number is a §11.4.201(6) false-null: its output reads as a census when it is a FLOOR. Anyone fencing an exclusion list

against that number (§11.4.224(E)) would write the fence against an undercount and lock the invisible sites out of scope permanently.

FOUR DISTINCT CAUSES, each independently falsifiable:

L1 — TRAILING COMMENT DEFEATS THE REGEX (10 of the 12). The pattern anchors the handler line with ...: `[[:space:]]*$`, so except Exception: `# noqa: S110` never matches. The irony is load-bearing: the sites a human consciously reviewed and annotated are exactly the ones the gate cannot see.

L2 — TUPLE CLAUSE DEFEATS THE REGEX (2 of the 12). The exception-type group is `[A-Za-z_\.]+`, which cannot match `(`, so except `(0SError, ValueError):` is invisible.

L3 — A COMMENT BETWEEN except AND pass DEFEATS THE BODY CHECK. The detector reads exactly `lineno+1`. Zero current instances, but demonstrated live. This one has a nasty second-order effect: a well-intentioned reviewer adding an explanatory comment INSIDE a handler makes that site vanish from the gate. The triage agent hit this itself — its first patch put comments inside two handlers, and the resulting count would have read 28 while only 6 sites were genuinely eliminated. It caught and corrected that rather than reporting the better number, which is exactly the §11.4 discipline working.

L4 — THE SHAPE ITSELF, not the regex. The body must be exactly pass, so except: `return <default>` is out of scope BY CONSTRUCTION. 13 such sites remain in boba (12 vendored `community/`, 1 `linuxtracker.py:51`), plus `plugins/rutor.py:121` (except: `return int(time.time())`) which a structural invariant found and which is now fixed. Returning a silent default is the SAME defect class as pass — arguably worse, because the caller receives a plausible value rather than nothing.

RECOMMENDED FIX: replace the text-matching detector with an AST-based one for Python. except handlers are trivially enumerable from `ast.Try.handlers`, and a handler whose body neither re-raises nor logs nor returns a distinguishable failure signal is decidable structurally. Text matching cannot reach L1-L4 without accumulating epicycles; each of the four above is a separate regex patch under the current design.

WHATEVER THE FIX, IT MUST BE FALSIFIABLE (§1.1): the paired mutation must include one fixture per cause — trailing comment, tuple clause, comment-before-pass, and `return <default>` — each of which the CURRENT detector passes and

a correct one must FAIL. A negative control is required too: a correctly-narrowed handler that logs and re-raises must NOT fire (§11.4.201(1)).

CONSUMER-SIDE NOTE (already fixed, boba):

pre_build_verification.sh invariant 39 counted the gate's own SUMMARY line as a finding, because that line also starts with the failure marker. Each failing root added exactly one phantom, so the reported total read 38 when the gate itself said 36. Fixed by matching the finding STRUCTURE (a finding names " at :"; a summary never does) rather than the marker glyph. That is a separate defect from the four above, in the consumer, and is NOT part of this item.

Affected scope / file-scope manifest: constitution/scripts/gates/cm_dangerous_combination_fail_closed.sh (+ its paired mutation test)

Reproduction / context: Run the gate over plugins/ and compare to an AST enumeration of ast.Try.handlers: gate=30, AST=42, 12 missed, 0 false positives. Each cause reproduces standalone: except Exception: # noqa (L1); except (OSError, ValueError): (L2); a comment between except and pass (L3); except: return

(L4).

Acceptance criteria: Detector finds all 42 AST-confirmed sites with zero false positives, with a paired §1.1 mutation carrying one fixture per cause (all four currently PASS the detector and must FAIL a correct one) plus a negative control that must NOT fire on a correctly-narrowed logging-and-re-raising handler.

BOB-148 — Standing red unit test nothing tracked: test_no_credentials asserts has_session False, gets True — real defect or non-hermetic test, undecided

Status: Fixed (→ Fixed.md) **Type:** Bug **Severity:** Medium
Created-By: Claude

Reported-Via: §11.4.202 reporting directive bug on 2026-08-20T16:28:57Z **Reported-By:** Claude

What (the report, verbatim): tests/unit/
test_auth_coverage.py:303
TestAllTrackersAuthStatus::test_no_credentials

```
assert result["trackers"]["qbittorrent"]["has_session"] is
False
E   assert True is False
```

The test asserts that with NO credentials configured,
qbittorrent reports has_session=False. It reports True.

PROVEN PRE-EXISTING, by experiment rather than by reasoning: a detached worktree at the session-start commit e335dde reproduces the identical failure. Nothing in this session touched download-proxy/src/api/auth.py or this test file (git log over the session range for both paths is empty). It was already red and nothing was tracking it — which is the actual defect worth recording: a standing red unit test that no item names will be re-discovered forever and attributed to whoever touches the tree next. It was in fact attributed twice today before being pinned down.

TWO HYPOTHESES, both plausible, NEITHER confirmed (§11.4.6 — do not pick one without evidence):

(H1) A REAL DEFECT: the auth-status path reports has_session=True on the strength of something other than a credential — a cached cookie, a default-constructed client, or a truthy default in the status assembler. If so, the operator-visible consequence is that the dashboard would show a tracker as authenticated when it is not.

(H2) A NON-HERMETIC UNIT TEST (§11.4.27(A)): the test reads real state rather than a stub, and the live qBittorrent container at :7185 — which is UP and healthy on this host — supplies a genuine session. Under that hypothesis the test would PASS on a host with the stack stopped, which makes it environment-dependent, i.e. FLAKY, and §11.4.248 quarantine territory rather than a product bug.

DECISIVE EXPERIMENT (cheap, and it distinguishes them in one run): execute this single test with the stack stopped, or with the qBittorrent host/port pointed at a closed port. If it PASSES, H2 holds and the fix is to make the unit test hermetic (mock the client) — the product is fine. If it still FAILS, H1 holds and the fix is in the auth-status assembly path.

Do NOT stop the stack casually to run this — other work depends on it. Prefer pointing the test at an unbound port via env override, which is reversible and affects nothing else.

NOTE the §11.4.226 evidence-class consequence: if H2 holds, this test has been asserting a RUNTIME condition from a unit-test layer all along, which is why it reads red on a developer host and would read green in a clean CI container — the worst possible polarity, since the environment that most resembles production is the one where the test stays silent.

Affected scope / file-scope manifest: tests/unit/
test_auth_coverage.py:303, download-proxy/src/api/auth.py
(auth-status assembly)

Reproduction / context: timeout 300 .venv/bin/python -m
pytest tests/unit/
test_auth_coverage.py::TestAllTrackersAuthStatus::test_no_c
redentials -q -import-mode=importlib -> assert True is False.
Reproduces identically in a detached worktree at e335dde
(session start).

Acceptance criteria: The H1/H2 experiment is run and recorded. If H2: the unit test is made hermetic (no live-stack dependency) and passes with the stack both up and down. If H1: the auth-status path no longer reports has_session without a credential, with a RED test capturing it first.

BOB-149 — Managed-plugin count diverges 43/42/48 across constitution, CLAUDE.md and the README badge; the badge is hand-maintained and unguarded

Status: Queued **Type:** Bug **Severity:** Low **Created-By:** Claude

Reported-Via: §11.4.202 reporting directive bug on 2026-08-20T22:35:42Z **Reported-By:** Claude

What (the report, verbatim): The managed-plugin count diverges three ways across the repo:

.specify/memory/constitution.md : 43 (and enumerates all 43 by name) CLAUDE.md : 42 (“42 managed plugins”)
README.md badge : 48 (plugins-48)

AUTHORITATIVE SOURCE, per the constitution’s own text (“the install-plugin.sh managed list is the canonical curated set”): the PLUGINS=() array in install-plugin.sh holds exactly 43 entries.

Counted with a control needle (§11.4.201(7)(b)): the extractor was verified to match a known member (“rutracker”) before its count was trusted. A first attempt returned 0 because the array entries are QUOTED (“academictorrents”) and the pattern was unquoted — a false-null caught by the needle rather than reported as “no plugins”.

So the constitution is CORRECT and the other two are the drifted copies.

Neither wrong number is harmless: - CLAUDE.md’s 42 is the file agents read as project instruction, so the wrong number is the one most likely to be propagated into new work. - README.md’s 48 matches NEITHER the curated array (43) NOR plugins/*.py (36) NOR plugins/**/*.py (69). It is not a stale-but-once-true number; nothing in the repo currently equals 48, so its provenance is unknown.

The badge is additionally UNGUARDED: compute-badges.sh does not derive the plugins badge at all — it is one of the hand-maintained ones. That is the same shape as the challenges/pre-build badges fixed at a6f36fa, which had been stale by 7 and 14 while the script printed “cross-checked, matches existing badge”. The plugins badge escaped that fix because it was never in the script’s scope.

Acceptance: 1. CLAUDE.md states 43, matching the authoritative array. 2. The README plugins badge is DERIVED by compute-badges.sh from the PLUGINS=() array (not hand-typed), so it cannot silently drift again. 3. tests/unit/test_compute_badges_all_badges_updated.sh is extended to cover the plugins badge, so the guard covers every machine-derived badge rather than the subset that happened

to be fixed first. 4. Verified in both directions (§11.4.201(1)): the guard FAILs when the badge disagrees with the array, and PASSes when they agree.

Filed rather than fixed inside the v1.4.0 constitution amendment so the documentation fix and the governance change stay independently reviewable.

Affected scope / file-scope manifest: CLAUDE.md, README.md (plugins badge), scripts/compute-badges.sh, tests/unit/test_compute_badges_all_badges_updated.sh

Reproduction / context: Count the PLUGINS=() array in install-plugin.sh (43, needle-verified) and compare against CLAUDE.md ('42 managed plugins') and the README badge (plugins-48).

Acceptance criteria: CLAUDE.md says 43; the README badge is derived by compute-badges.sh from the array; the badge guard covers it; both polarities verified.

BOB-150 — pre_build_verification.sh invariant labels read N/44 but only 35 invariants are labelled

Status: Queued **Type:** Task **Severity:** Low **Created-By:** AI

Reported-Via: §11.4.202 reporting directive task on 2026-08-21T14:46:07Z **Reported-By:** AI

What (the report, verbatim): The pre-build runner announces each invariant as [N/44], but only 35 invariants carry a label and only 35 distinct CM-* gate names are announced. Numbers 33-38, 40, 42 and 43 are unused; there are no duplicates. An operator reading the output sees '[44/44]' scroll past and reasonably concludes 44 invariants ran, when 35 did - the output overstates coverage by 9. This is PRE-EXISTING and was surfaced while wiring CM-OWNERSHIP-INVARIANTS, which deliberately took free slot 33 precisely to avoid a 35-label renumbering that would have conflicted with concurrent work. That gate neither

introduced nor fixed this. Filing rather than absorbing it silently, per the closed-or-tracked rule: an unstated finding reads as no finding.

Affected scope / file-scope manifest: scripts/
pre_build_verification.sh

Reproduction / context: `grep -oE '[[0-9]+/44]' scripts/
pre_build_verification.sh | wc -l -> 35` (denominator says 44);
numbers 33-38, 40, 42, 43 are unused, no duplicates. Distinct
CM-* names announced in the file: 35, which agrees with the
label count and not with the denominator.

Acceptance criteria: Either the denominator matches the
real number of labelled invariants, or the numbering is
compacted to be contiguous - and whichever is chosen, a
gate or the runner itself derives the denominator rather than
restating it as a literal, so the two cannot drift apart again.

BOB-151 — CM-SCRIPT-DOCS-SYNC is a named gate with no implementation, and 24 of 36 scripts have no companion doc

Status: Queued **Type:** Task **Severity:** Medium **Created-By:**
AI

Reported-Via: §11.4.202 reporting directive task on
2026-08-21T15:05:33Z **Reported-By:** AI

What (the report, verbatim): The §11.4.18 script-
documentation rule names a gate CM-SCRIPT-DOCS-SYNC,
but no executable file in this repository implements it.
Because nothing measures the obligation, 24 of 36 scripts
under scripts/ have no docs/scripts/.md companion - two
thirds of them. This is the named-gate ledger gap the
constitution itself warns about: a gate that is named but
never implemented reads as coverage while providing none,
and the corpus grows words instead of enforcement.
Surfaced while writing the two ownership companions, which
were themselves only written because the feature plan asked
for them explicitly - not because any gate demanded it. Filing

rather than absorbing: an unstated finding reads as no finding, and a 67% documentation gap that nothing reports will not shrink on its own.

Affected scope / file-scope manifest: `scripts/.sh`, `docs/scripts/.md`, `scripts/pre_build_verification.sh`

Reproduction / context: `grep -rl CM-SCRIPT-DOCS-SYNC -include='.sh' -include='.py' . (excluding submodules) -> 0` files. Control needle in the same command shape: `CM-OWNERSHIP-INVARIANTS -> 3` files, `CM-KILLPG-PGID-GUARD -> 7` files, so the search is not blind. Then: `for s in scripts/*.sh; do [ -f docs/scripts/$(basename $s .sh).md ] || echo missing; done -> 24 of 36 missing.`

Acceptance criteria: Either the gate is implemented and the 24 missing companions are written (the count becoming a monotone-decreasing ratchet so it cannot grow), or the obligation is explicitly scoped down to a defined subset with the reason recorded. What is NOT acceptable is the current state, where the rule is stated and nothing measures it.

BOB-152 — Constitution sweep walks vendored third-party code: 82% of one gate's 38,291 findings come from submodules/

Status: Queued **Type:** Bug **Severity:** Medium **Created-By:** AI

Reported-Via: §11.4.202 reporting directive bug on 2026-08-21T15:10:28Z **Reported-By:** AI

What (the report, verbatim): The constitution sweep passes `-root` at the repository root, so every gate walks vendored third-party code the project neither wrote nor ships: `submodules/helixqa/tools/opensource/perfetto`, `chroma`, `skyvern`, `mem0` and so on. Measured today, 82% of one gate's findings originate there. This is a false-positive refusal at scale: it fails the sweep over code that cannot be fixed here, and it buries the 4497 first-party findings that might actually matter under 31496 that do not. A second, subtler instance: `cm_killpg_pgid_guard` flags its OWN golden-

bad fixtures and the BOB-126 incident docstrings - the gate reporting on the very artifacts that prove it works. Surfaced while fixing the .worktrees/ half of this problem (BOB-143), which was the smaller 6% slice; that fix was deliberately scoped to .worktrees/ rather than quietly widened to cover this, because excluding submodules/ is a materially larger decision about what the sweep is for and deserves its own review.

Affected scope / file-scope manifest: scripts/verify-all-constitution-rules.sh, config/constitution-sweep.conf, constitution/scripts/gates/*

Reproduction / context: config/constitution-sweep.conf line 27 passes 'DEFAULT -root @ROOT@', so every gate walks the whole repository. Per-tree census of
cm_oracle_strategy_named_and_independent over the full root: 38291 findings total - 31496 (82%) from submodules/, 2280 (6%) from .worktrees/, 4497 from the real tree.
cm_killpg_pgid_guard: 18 findings - 9 from submodules/, and of the 8 in the real tree several are the gate's OWN golden-bad fixtures and BOB-126 docstrings.

Acceptance criteria: The sweep scans code this project actually ships. Vendored third-party trees under submodules/ are excluded or scoped explicitly, and any gate's own golden-bad fixtures are excluded from its own scan - with the exclusion validated in BOTH directions (a planted violation in first-party code must still FAIL) so this does not become narrow-until-green.

BOB-153 — Go profile cannot build: go.mod requires go 1.26.2 but the Dockerfile builder is golang:1.23-alpine

Status: Fixed (→ Fixed.md) **Type:** Bug **Severity:** Medium
Created-By: AI

Reported-Via: §11.4.202 reporting directive bug on 2026-08-21T15:40:00Z **Reported-By:** AI

What (the report, verbatim): The Go backend profile is unbuildable. qBitTorrent-go/go.mod declares 'go 1.26.2' while qBitTorrent-go/Dockerfile builds with 'FROM golang:1.23-alpine', so the build aborts before compiling anything. Found while attempting feature 002 quickstart Scenario 6, which exercises the go profile to confirm the ownership fix covers every service (FR-016) - the scenario could not run at all, for a reason unrelated to ownership. Two consequences worth separating. First, this is a plain build defect and is pre-existing. Second, and more awkward, it means FR-016 coverage for the go profile currently rests on surface-equivalent measurement (its Dockerfile's final stage is alpine:3.19 with no USER directive, so it runs as container root and inherits the correct rootless mapping) rather than on a live running service. That reasoning is sound but it is not the same evidence as a probe against a real container, and it is recorded as the weaker evidence it is. Also noted while investigating: the go profile has no Hard-Stop-#3-compliant invocation path - start.sh has no -profile flag, so the only route is a raw compose command, and it would collide on port 7187 with the running Python proxy since both use network_mode host.

Affected scope / file-scope manifest: qBitTorrent-go/go.mod, qBitTorrent-go/Dockerfile

Reproduction / context: grep '^go' qBitTorrent-go/go.mod -> 'go 1.26.2'; grep 'FROM golang' qBitTorrent-go/Dockerfile -> 'FROM golang:1.23-alpine AS builder'. Building the go profile fails: 'go: go.mod requires go >= 1.26.2 (running go 1.23.12; GOTOOLCHAIN=local)'.

Acceptance criteria: The go profile builds. Either the builder image is raised to a toolchain satisfying go.mod, or go.mod's directive is lowered to what the builder provides - and whichever is chosen, a check ties the two together so they cannot drift apart again, because nothing currently compares them.

BOB-154 — Host venv and production container run different starlette versions (1.4.1 vs 1.6.0)

Status: Ready for testing **Type:** Bug **Severity:** Medium
Created-By: AI

Reported-Via: §11.4.202 reporting directive bug on
2026-08-21T15:46:07Z **Reported-By:** AI

What (the report, verbatim): The host virtualenv used to run this project's unit tests resolves starlette 1.4.1 while the running qbittorrent-proxy container resolves 1.6.0. Every other implicated package matches, so this is a genuine unpinned-transitive-dependency drift rather than a deliberate difference. It matters because it silently weakens every test result: a green suite on the host is evidence about 1.4.1, and production is 1.6.0. A behavioural change between those versions would be invisible to the tests that exist to catch it, which is the build-once-run-the-same-bytes property the constitution asks for. Surfaced while investigating BOB-129, where the defect happened to reproduce IDENTICALLY at both versions - which is what allowed that ticket's slowapi/starlette-incompatibility premise to be refuted. That was luck, not design: the next divergence may not be version-independent, and nothing would tell us.

Affected scope / file-scope manifest: download-proxy/requirements.txt, .venv, container qbittorrent-proxy

Reproduction / context: .venv/bin/python -c 'import starlette;print(starlette.__version__)' -> 1.4.1 ; podman exec qbittorrent-proxy python -c 'import starlette;print(starlette.__version__)' -> 1.6.0. Same fastapi (0.141.1), same slowapi (0.1.10), same limits (5.8.0) - starlette alone diverges.

Acceptance criteria: The interpreter that runs the tests and the interpreter that serves production resolve the same versions, pinned so they cannot drift apart silently - or, if a divergence is deliberate, it is declared and a check asserts the declared pair rather than leaving it to chance.

BOB-155 — workable-items diff reports 'DB and Markdown are in sync' having opened zero Markdown files when -issues/-fixed are omitted

Status: Fixed (→ Fixed.md) **Type:** Bug **Severity:** High
Created-By: AI

Reported-Via: §11.4.202 reporting directive bug on
2026-08-21T15:57:00Z **Reported-By:** AI

What (the report, verbatim): The flagless form of the sync checker is a false-null generator. Called without -issues/-fixed it prints the same reassuring 'DB and Markdown are in sync' it prints after a real successful comparison, while having read no Markdown whatsoever. A blind instrument and a genuinely clean tree return the identical quiet verdict, which is precisely the failure class the constitution's measurement-integrity rules exist to prevent - and here it lives inside the project's own sync-verification tool. This bit for real today: the flagless form was used to VERIFY a reconciliation of five tracker rows and reported 'in sync'. The reconciliation happened to be correct - re-checked afterwards with the path-ful form, which is clean on the current tree and correctly reports 2 differences against a planted divergence - so the conclusion was right and the evidence for it was worthless. Nobody would have noticed, because the output is indistinguishable from a real pass. Surfaced by the BOB-136 investigation. Not fixed there because the file lives in the constitution submodule, which carries its own review and commit discipline and was dirty with concurrent work at the time. The caller-side exposure was closed instead (a gate now asserts zero flagless callers), but the engine itself still ships the trap for every other consumer of that submodule.

Affected scope / file-scope manifest: constitution/scripts/workable-items/cmd/workable-items/sync.go

Reproduction / context: Plant a real divergence: change a **Status:** line in docs/Issues.md only. Then: workable-items diff -db docs/workable_items.db -> 'diff: DB and Markdown are in sync' (WRONG - it compared nothing). The same command WITH paths: workable-items diff -db docs/workable_items.db -issues docs/Issues.md -fixed docs/Fixed.md -> '2 difference(s)' (correct). Restored byte-identical after the test.

Acceptance criteria: Omitting -issues/-fixed either REFUSES with a non-zero exit naming the missing input, or defaults to the conventional paths and says which files it read. What must never happen again is a confident 'in sync' verdict from a comparison that opened no Markdown at all - the verdict must always name its inputs so a reader can tell a real check from a vacuous one.

Closed 2026-08-21 — constitution commit 16b67b0, pushed to 8 upstreams. Chose REFUSE plus an explicit --db-only opt-

in, because the alternative (defaulting to conventional paths) is NOT implementable in a shared submodule without baking one consumer's docs/Issues.md layout into it. Sibling precedent settled it: sync md-to-db and sync db-to-md in the same file ALREADY refuse when every path flag is absent, so diff was the exception. Every verdict now NAMES ITS INPUTS — “compared 152 Markdown item(s) against 152 DB item(s); read

/Issues.md,

/Fixed.md” — which is the property that makes the failure class impossible rather than merely unlikely.

ROOT CAUSE WITH HISTORY: this defect was INTRODUCED BY THE FIX FOR ITS OWN MIRROR IMAGE. Before 2026-08-10 the flagless form ran the absent-in-Markdown loop against an EMPTY parsed set and flagged every DB row — a FAIL-bluff. The fix added a haveMarkdown gate that correctly silenced the noise, then fell through to the unconditional success verdict. Same seam, opposite polarity: a FAIL-bluff traded for a PASS-bluff. Worth recording, because “we fixed the false positives” is exactly how a false negative gets installed.

TWO ADJACENT DEFECTS SURFACED, both worse than the one filed: (1) BOB-155 was ALREADY being caught by a test's GREEN branch — a standing red nobody saw because the suite is evidently never run in GREEN mode, a coverage escape where the check existed and nothing executed it; (2) the suite was RED IN BOTH POLARITY MODES beforehand, and one test's GREEN branch had begun ASSERTING THE DEFECT. Both reconciled to assert the new mechanism rather than fake-passed or reverted. One lesser instance fixed in passing: md-to-db printed the hardcoded label Issues.md: even when reading a renamed tracker — a verdict misnaming its input AND a baked-in filename in a shared submodule.

Siblings audited EMPIRICALLY, not assumed: md-to-db and db-to-md refuse; validate has no optional inputs and names its item count, so it cannot be blind. Commit seam verified against a purpose-built PRE-FIX binary and the fixed one — identical behaviour, because all three of its checks already pass -issues/-fixed. Zero project literals among added lines, control-needed. ## BOB-156 — BOB-145 event-loop regression guard is load-sensitive and flaky: 8786ms under host load vs a 900-1500ms ceiling calibrated on a quiet host

Status: Queued **Type:** Bug **Severity:** Medium **Created-By:** AI

Reported-Via: §11.4.202 reporting directive bug on 2026-08-21T17:01:20Z **Reported-By:** AI

What (the report, verbatim): The regression guard added with the BOB-145 fix asserts an event-loop block ceiling calibrated on a quiet host. Under real host contention the block window scales with BOTH N and load, so the same code that passes at 349ms median can measure 8786ms - worse than the pre-fix number the test exists to detect. That makes it FLAKY, and a flaky test is corrosive in a specific way this project has already recorded: every ignored red trains everyone to dismiss the next one, so a real regression eventually gets waved through as 'that one again'. Two directions are wrong: raising the ceiling until it stops failing would blind it to the defect, and leaving it flaky poisons every future run. Found while verifying BOB-137 against the live service.

Affected scope / file-scope manifest: tests/unit/merge_service/test_dedup_event_loop_blocking.py

Reproduction / context: Under host load 18-24 on 8 cores (concurrent agents), the same N=400 merge froze the event loop for 8786ms - WORSE than the 3970ms pre-fix figure the ceiling was calibrated against. 2 fail / 2 pass across four runs.

Acceptance criteria: The guard gives the same verdict on a loaded host as on a quiet one - or it measures something contention-independent. A threshold that only holds when nothing else is running is not a regression guard, it is a weather report.

BOB-157 — Our own BOB-137 stall watchdog can segfault the merge service: fault handler dump_traceback(all_threads=True) hits an unpatched CPython 3.12 defect

Status: Fixed (→ Fixed.md) **Type:** Bug **Severity:** High
Created-By: AI

Reported-Via: §11.4.202 reporting directive bug on
2026-08-21T17:18:18Z **Reported-By:** AI

What (the report, verbatim): Our own diagnostic is a crash vector, and it is still armed. download-proxy/src/main.py:135 calls `faulthandler.dump_traceback(file=sink, all_threads=True)` on a live 18-thread process; three further registrations at 214/217/219 use `all_threads=True` as well. Upstream python/cpython#116008 and #128400 are the same NULL `f_executable` in `dump_frame()` at `Python/traceback.c:1190` - FIXED and backported to 3.13/3.14, with NO 3.12 backport listed. The container ships Python 3.12.13. Container logs still show 'BOB-137 stall watchdog armed: stall>20.0s'. It has not fired again only because BOB-137's root cause was improved enough that the loop rarely stalls past the threshold - but BOB-137 is NOT closed, and its live verification the same day measured the loop still blocking, with 18.4% of probes stalled 1-5s and two dead events in 15 minutes. So this is latent, not resolved: one 20s stall away. Note the perverse shape - the worse the wedge gets, the more likely the tool built to diagnose it is to kill the process, destroying the evidence it exists to capture. Found while investigating BOB-131, whose own premise (a common crash) turned out to be false: no common process crashed; the ticket conflated this python3 segfault with an unrelated 14-hour absence caused by a host power-off.

Affected scope / file-scope manifest: download-proxy/src/main.py:135 (and the registrations at 214/217/219)

Reproduction / context: 2026-08-20 17:56:26 CEST, one occurrence: kernel 'python3[314359]: segfault at 70 ... in libpython3.12.so.1.0' plus the container's own truncated dump ending mid-' File '. The kernel Code: bytes at IP were byte-matched against the library inside the running container (MATCH), decoding to `mov r14,[r12]` (`frame->f_executable = NULL`) then `mov rax,[r14+0x70]` (`code->co_filename`) -> fault; the preceding `lea` resolves to the literal ' File ' with `edx=7`, its exact length. The watchdog had fired 17 times in 16 minutes that day; dump 17 completed to the file sink, then the stderr pass crashed.

Acceptance criteria: The diagnostic cannot crash the service it diagnoses. Either `all_threads=True` is dropped for the periodic dump, or the dump is gated behind something

that cannot fault on a live multi-threaded process, or the runtime moves to a Python where the upstream fix is present - and whichever is chosen, the choice is recorded against the upstream issue so a later runtime bump does not silently re-arm it.

BOB-158 — tests/conftest.py cannot run on the production interpreter: binds asyncio.events._get_event_loop_policy, a 3.13+ private API, while production is 3.12.13

Status: Queued **Type:** Bug **Severity:** High **Created-By:** AI

Reported-Via: §11.4.202 reporting directive bug on 2026-08-21T17:29:10Z **Reported-By:** AI

What (the report, verbatim): The test suite has silently acquired a dependency on an interpreter production does not run. tests/conftest.py:207 calls `asyncio.events._get_event_loop_policy()` - a PRIVATE API that does not exist before Python 3.13 - in an AUTOUSE fixture, so it affects every test. The host venv is CPython 3.14.6; the container is 3.12.13, pinned deliberately in docker-compose.yml as `python:3.12-alpine` and independently declared twice more in pyproject.toml (ruff target-version py312, mypy python_version 3.12). So the venv contradicts the project's own declared target by documentary evidence, not opinion. THIS IS NOT A THEORETICAL GAP: every green suite run to date is evidence about 3.14.6 while users are served 3.12.13, and this specific incompatibility means the suite CANNOT run on the production interpreter at all - it produces 870 teardown errors. It went unnoticed because nobody had ever run the suite on 3.12. Found while fixing BOB-154 (dependency drift), whose reconciliation is BLOCKED on this: rebuilding the venv on 3.12 is the fix for the drift, and doing so makes the suite unrunnable until this is resolved. A candidate patch shape was verified in isolation on BOTH 3.12.13 and 3.14.6 under -W error::DeprecationWarning, but not as an integrated change - and another stream was editing conftest.py concurrently, so it was deliberately not applied.

Affected scope / file-scope manifest: tests/conftest.py:207
(_cleanup_event_loop fixture)

Reproduction / context: .venv/bin/python -c 'import
asyncio.events as e;
print(hasattr(e, "_get_event_loop_policy"))' -> True (3.14.6).
podman exec qbittorrent-proxy python -c same -> False
(3.12.13). Running the suite on 3.12.13 produces 870
teardown errors. A second incompatibility in the same
fixture: policy.get_event_loop() raises DeprecationWarning on
3.12, which this project's pytest config turns into an error.

Acceptance criteria: The suite runs on the interpreter
production runs. Both incompatibilities in
_cleanup_event_loop are resolved, verified on 3.12.13 under
-W error::DeprecationWarning, and the venv is rebuilt on
3.12 so every subsequent test result is evidence about the
deployed runtime.